

Ahmed Yosri Hassan

CONSULTING ENGINEER, MARINE STRUCTURES

I design and verify ship structures — from polar icebreakers to cargo vessels — with a research background in fatigue of marine structures and ship dynamics. Ten years across FEA, ship dynamics, intact and damage stability and structural design.

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English · Arabic · Finnish (learning)

View the online interactive version of this CV:

https://ahmedyosriu3-dev.github.io/CV_ST/

4.88 / 5	**Dean's List**	**1st / 51**	**3 papers**
M.Sc. Aalto, with honours	Aalto School of Engineering 2020–21	B.Sc. class valedictorian	incl. Ocean Engineering — 1st-ranked marine journal

CAREER

Elomatic Consulting & Engineering — Helsinki

MAY 2022 — PRESENT

- **Consulting Engineer — Marine Structural Design & FEA** · MAR 2025 — PRESENT
- **Senior Structural Analyst & Designer** · MAY 2023 — MAR 2025
- **Structural Analyst & Design Engineer** · MAY 2022 — MAY 2023

- Global and local strength, vibration, harmonic and buckling FEA; design optimization of machinery and crane foundations, deckhouse, working decks, masts and moon pool.
- Ship dynamics and mooring analysis, global dry and wet eigenmodes.
- Hull girder strength and scantling with class software (DNV Nauticus Hull, LR ShipRight); 3D ship structure design in SpaceClaim and NAPA Steel.
- Intact and damaged stability for concept designs and loading scenarios; offshore wind turbine research and patent design development.

Aalto University — Marine and Arctic Technology

JAN 2021 — MAY 2022 · ESPOO / HELSINKI

- **Research Assistant** (Jan–May 2022) — master-thesis research: ship design, hydroelasticity, ship dynamics, structural analysis, acoustic response, ship hydrodynamics.
- **Teaching Assistant** (Jan 2021 – Jan 2022) — Ship Dynamics, Principles of Naval Architecture and Ship Policy courses.

Alexandria University, Faculty of Engineering — Teaching & Research Assistant

FEB 2018 — DEC 2020 · EGYPT

Master's research on fatigue assessment of aged marine structures — plus lab sessions and course design for the Naval Architecture and Offshore & Ocean Engineering programmes.

In parallel — freelance engineering (2016–2020):

- **Project Engineer, United Group Enterprises** — ship launching and pulling-up with marine airbags (barges and Nile floating hotels); modular barge and buoy design, keel block plans, underwater surveys; FEA studies.
- **Marine Project Engineer, Swiss Logistics Consulting GmbH** — FEA, ship dynamic response and heavy lifting for clients DB Schenker, Linde plc and Minerals Technologies.
- **Freelance naval architecture & mechanical design** — Upwork and Freelancer.com, clients worldwide.

Summer training — shipyards & maritime authorities

2013 — 2015 · ALEXANDRIA, EGYPT · SEVEN INTERNSHIPS, 1–3 MONTHS EACH

Alexandria Shipyard · Alexandria Port Authority · Egyptian Ship Repairs and Building Company · Egyptian Marine Supply and Contracting Company (CONSUP) · Egyptian Authority for Maritime Safety · DODO RORO ship · Aida 4 (AASTMT).

TECHNICAL EXPERTISE

Structural analysis & design

Global strength FEA — hull girder still-water, wave and dynamic wave loads; class-approved functional design for two icebreakers; in-house ANSYS load application and result-check tools.

Fatigue — responsible for global and local fatigue on Polarstern II and the Canadian polar icebreaker; icebreaker fatigue method based on IACS rules.

Local FEA — engine, crane and mooring foundations, double bottom, masts, working decks, moon pool and welds.

Vibration & harmonic — mast vibration optimized against machinery and propulsion excitation; dry and wet hull girder eigenmodes; acceleration and comfort checks.

Buckling & nonlinear — FEA verified with hand calculations; ultimate strength against grounding loads in research.

Naval architecture & stability

Stability — intact and damaged stability, compartment modelling and tank plans; step-by-step loading scenarios for heavy-lifting projects.

Scantling & arrangement — hull girder scantling with DNV Nauticus Hull and LR ShipRight; general arrangement for concept ships.

Marine operations — launching and pulling-up with marine airbags for barges and floating hotels up to 110 m.

Ship dynamics & hydrodynamics

Wave loads & motions — RAOs, motions and global wave loads with 3D potential flow (AQWA, HydroSTAR); frequency- and time-domain response.

Mooring, towing & hydroelasticity — mooring and towing loads mapped onto the structure; hydroelastic response (Aalto research); powering with Holtrop; basic CFD.

AI-assisted coding for marine tasks

Tools — in-house apps for load application, result verification and parametric fatigue sheets; DayBook and Finnish Tutor published on the Microsoft Store; MATLAB spectral fatigue code for M.Sc. research.

SELECTED PROJECTS

Polarstern II — German polar research icebreaker (Elomatic) — basic and functional design.

Canadian Polar Icebreaker — Seaspan Vancouver (Elomatic) — basic, functional and production design.

RoPax ferry concept design (Elomatic) — concept design.

Ammonia-ready LNG-fuelled Capesize bulk carrier & VLCC (NYK & MTI — ARLFV) — concept phase design.

Float Foundation — offshore installation patents (Elomatic R&D).

Heavy-lifting deck carrier (Elomatic).

SOFTWARE

STRUCTURE ANSYS Mechanical (modal, static and dynamic structural, transient, nonlinear, harmonic, modal acoustic; nCode fatigue; command scripting) · SpaceClaim · DNV Nauticus Hull & LR ShipRight · AutoCAD · SolidWorks · Cadmatic Hull · ABAQUS · FEMAP

STABILITY NAPA (Steel, compartments and tanks, meshing, intact and damage stability, seakeeping, powering, hull modelling and fairing) · Maxsurf

DYNAMICS BV HydroSTAR & Homer · ANSYS AQWA · ANSYS Modal Acoustic · ANSYS Fluent & CFX

CODING AI-assisted coding (Dart, Python, C++, HTML) · Excel (parametric sheets and macros) · MATLAB · MathCad

ALSO Microsoft Office · photo and video editing

EDUCATION

M.Sc. (Technology) — Mechanical Engineering, Marine Technology & Solid Mechanics

Aalto University, School of Engineering, Finland · 2020–2022

GRADE AVERAGE 4.88/5 · COMPLETED WITH HONOURS · DEAN'S LIST 2020–2021 · THESIS 5/5

Thesis: Prediction of global loads of intact, damaged and grounded ships using Fluid–Flexible Structure Interaction methods. Supervisor: Prof. Spyros Hirdaris.

M.Sc. (Research) — Marine Engineering & Naval Architecture

Alexandria University, Faculty of Engineering, Egypt · 2017–2020

COURSE CGPA 3.708 / 4.0

Thesis: An Engineering Assessment for Fatigue Life of Corroded Ship Structure.

B.Sc. — Marine Engineering & Naval Architecture

Alexandria University, Faculty of Engineering, Egypt · 2011–2016

85.51% — EXCELLENT WITH DEGREE OF HONOUR · RANKED 1ST OF 51 — CLASS VALEDICTORIAN

PUBLICATIONS

Accumulated fatigue damage assessment of side structural details in a double hull tanker based on spectral fatigue analysis approach

Ocean Engineering 251 (2022) 111069 — 1st-ranked journal in the marine field. A. Yosri, H. Leheta, S. Saad-Eldeen, A. Zayed.
doi.org/10.1016/j.oceaneng.2022.111069

Influence of stress concentration on fatigue life of corroded specimens under uniaxial cyclic loading

Alexandria Engineering Journal 60.6 (2021) 5205–5216. A. Yosri, A. Zayed, S. Saad-Eldeen, H. Leheta.
doi.org/10.1016/j.aej.2021.04.004

Fatigue Assessment of Aged Steel Specimens Under Uniaxial Cyclic Loading

Fracture, Fatigue and Wear (pp. 523–532), Springer, 2020. A. Yosri, A. Zayed, S. Saad-Eldeen, H. Leheta.

BEYOND THE DRAWING OFFICE

I am a deeply practical person and I love DIY. I perform all the regular maintenance on my car. The same passion extends to cycling: I custom-built a carbon-fibre gravel bike from the frame to the gears, sourcing and assembling every component myself. Whether on a multi-day bikepacking trip or in my daily fitness routine, I am driven by the same curiosity and attention to detail that I bring to the marine field — these hobbies are an extension of my engineering mindset, not just leisure.

INTERACTIVE CV — WITH THE EXPERTISE EXPLORER, PROJECT MEDIA AND CERTIFICATES — AVAILABLE ONLINE · https://ahmedyosriu3-dev.github.io/CV_ST/